

Recap

$T: V \rightarrow W$ linear between finite dimensional vector spaces.

$\beta = \{v_1, \dots, v_n\}$, $\gamma = \{w_1, \dots, w_m\}$ bases

$v = a_1 v_1 + \dots + a_n v_n \rightarrow \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \leftarrow$ coordinate expression for a vector
 $= [v]_{\beta} \in \mathbb{R}^n$

$T(v_j) = a_{1j} w_1 + \dots + a_{mj} w_m \rightarrow [T(v_j)]_{\gamma} = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix} \in \mathbb{R}^m$
 $[T]_{\gamma}^{\beta} = (a_{ij}) = ([T(v_1)]_{\gamma}, \dots, [T(v_n)]_{\gamma})$

matrix representation of T wrt β, γ

Claim. $A \in M_{m \times n} \rightarrow L_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$
 $\bar{x} \mapsto A\bar{x}$

If β and γ are the standard bases for $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively,
 then $A = [L_A]_{\gamma}^{\beta}$

Linear transforms:

$\mathbb{R}^n$ standard basis $\{e_1, \dots, e_n\}$

Thm $T: V \rightarrow W$, $\beta, \gamma \leftarrow$ fixed bases

for all $v \in V$, $[T(v)]_{\gamma} = [T]_{\gamma}^{\beta} [v]_{\beta}$ (T acts like matrix multiplication)

Pf $\beta = \{v_1, \dots, v_n\}$, $\gamma = \{w_1, \dots, w_m\}$

$$\begin{aligned} [T(v)]_{\gamma} &= [T(a_1 v_1 + \dots + a_n v_n)]_{\gamma} \\ &= [a_1 T(v_1) + \dots + a_n T(v_n)]_{\gamma} \\ &= a_1 [T(v_1)]_{\gamma} + \dots + a_n [T(v_n)]_{\gamma} \end{aligned}$$

$W \rightarrow \mathbb{R}^m$ is linear, $T(v_i) \in W$.

$$\begin{aligned} w &\mapsto [w]_{\gamma} \\ &= ([T(v_1)]_{\gamma}, \dots, [T(v_n)]_{\gamma}) \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \\ &= [T]_{\gamma}^{\beta} [v]_{\beta} \end{aligned}$$

Vector space of linear transforms

$$L(V, W) = \{T: V \rightarrow W \mid T \text{ linear}\}$$

Corollary $L(V, W)$ is a vector space

Thm $L(\mathbb{R}^n, \mathbb{R}^m) \cong M_{m \times n}$

$$\begin{aligned} T \circ S &: X \rightarrow Z \\ x &\mapsto T(S(x)) \end{aligned}$$

Thm $T \circ S$ is linear ($T \circ S \in L(X, Z)$)

Matrix representation of Composition

let X, Y, Z be finite dimensional with bases $\alpha = \{x_1, \dots, x_n\}$,
 $\beta = \{y_1, \dots, y_m\}$, $\gamma = \{z_1, \dots, z_p\}$

Define $[T \circ S]_{\gamma}^{\alpha}$ in relation $[S]_{\beta}^{\alpha}$ and $[T]_{\gamma}^{\beta}$

Given $A \in M_{m \times n}$ and $B \in M_{n \times p}$, define $AB \in M_{m \times p}$ by
 $(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$

Thm $[T \circ S]_{\gamma}^{\alpha} = [T]_{\gamma}^{\beta} [S]_{\beta}^{\alpha}$

$$\begin{aligned} \text{Ex } A \in M_{m \times n} &\rightarrow L_A: \mathbb{R}^n \rightarrow \mathbb{R}^m \\ B \in M_{n \times p} &\rightarrow L_B: \mathbb{R}^p \rightarrow \mathbb{R}^n \end{aligned}$$

$$\begin{aligned} [L_B \circ L_A]_{\alpha}^{\gamma} &= [L_B]_{\beta}^{\gamma} [L_A]_{\alpha}^{\beta} \\ &= BA = [L_{BA}]_{\alpha}^{\gamma} \end{aligned}$$

because standard basis

$$L_B \circ L_A = L_{BA}$$

Pf $\alpha = \{x_1, \dots, x_n\}$ column by column computation

$$[T \circ S]_{\gamma}^{\alpha} = ([T \circ S](x_1))_{\gamma} \dots ([T \circ S](x_n))_{\gamma}$$

$$\begin{aligned} \text{the matrix representation of } T \text{ acting on the coordinate expression of } S(x_i) \text{ wrt } \beta. &= [T(S(x_1))]_{\gamma} \dots [T(S(x_n))]_{\gamma} \\ &= [T]_{\gamma}^{\beta} [S(x_1)]_{\beta} \dots [T]_{\gamma}^{\beta} [S(x_n)]_{\beta} \\ &= [T]_{\gamma}^{\beta} [S(x_1)]_{\beta} \dots [S(x_n)]_{\beta} \\ &= [T]_{\gamma}^{\beta} [S]_{\beta}^{\alpha} \end{aligned}$$

$$\begin{aligned} \text{Ex. } T_d: P_3 &\rightarrow P_2 \\ f &\mapsto f' \end{aligned}$$

$$\begin{aligned} T_i: P_2 &\rightarrow P_3 \\ f &\mapsto \int_0^x f(x) dx \end{aligned}$$

For standard bases:

$$[T_d]_{\beta}^{\gamma} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \quad [T_i]_{\gamma}^{\beta} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix}$$

$$[T_i]_{\gamma}^{\beta} [T_d]_{\beta}^{\gamma} = [T_i \circ T_d]_{\gamma}^{\beta} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{aligned} (T_i \circ T_d)(a_0 + a_1 x + a_2 x^2 + a_3 x^3) &= a_1 x + a_2 x^2 + a_3 x^3 \end{aligned}$$

coordinate representation

Matrix multiplication

Properties

(i) $A(BC) = (AB)C$ associative

(ii) $A(B+C) = AB+AC$ distributive

(iii) Not commutative. $AB \neq BA$ in general

Proof $A(B+C) = AB+AC$

Identity matrix and Kronecker delta

$$I_n = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

$n \times n$ identity matrix

$$(I_n)_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{otherwise} \end{cases}$$

Kronecker delta

Inverse linear transformation

For $A \in M_{n \times n}$, $A I_n = A$ and $I_n A = A$ (preserves identity)

Thm let β be a finite basis for V .

$$[I_V]_{\beta}^{\beta} = I_n, \text{ where } n = \dim(V)$$

Def An inverse of $T: V \rightarrow W$ is a map $S: W \rightarrow V$ such that

$$S \circ T = I_V \text{ and } T \circ S = I_W$$

Ex $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$S: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$(x, y) \mapsto (y, -x) \text{ has an inverse } (x, y) \mapsto (-y, x)$$

$$S \circ T(x, y) = S(y, -x) = (x, y)$$

$$T \circ S(x, y) = T(-y, x) = (x, y)$$

Thm Let $T: V \rightarrow W$ be linear.

1. T has an inverse iff T is 1-1 ($\text{NCT} = \{\vec{0}_V\}$) and onto ($\text{RCT} = W$)
2. If T has an inverse, then it is unique
3. T^{-1} is linear.

Prove that T^{-1} is linear:

$T: V \rightarrow W$ has inverse $T^{-1}: W \rightarrow V$

$$\begin{aligned} T^{-1}(w_1 + cw_2) &= T^{-1}(T(v_1) + cT(v_2)) = J_V(v_1 + cv_2) \\ &= v_1 + cv_2 = T^{-1}(w_1) + cT^{-1}(w_2) \end{aligned}$$

Image of basis by invertible linear map

Thm Suppose $T: V \rightarrow W$ is linear and invertible. If β is a basis for V , then $T(\beta)$ is a basis for W .

Pf let $\dim(V) = n$ and $\beta = \{v_1, \dots, v_n\}$ of V

$$T(\beta) = \{T(v_1), \dots, T(v_n)\}$$

1. Need $\text{span}(T(\beta)) = W$ and $w = a_1 T(v_1) + \dots + a_n T(v_n)$

$$\text{let } w = T(v)$$

$$v = a_1 v_1 + \dots + a_n v_n.$$

$$w = T(a_1 v_1 + \dots + a_n v_n) = a_1 T(v_1) + \dots + a_n T(v_n) \quad \checkmark \text{ done}$$

2. Need $T(\beta)$ is linearly independent.

$$\text{Suppose } a_1 T(v_1) + \dots + a_n T(v_n) = \bar{0}_W.$$

$$\text{Need } a_1, \dots, a_n = 0$$

$$T(a_1 v_1 + \dots + a_n v_n) = \bar{0}_W. \text{ Since } T \text{ is 1-1, } a_1 v_1 + \dots + a_n v_n = \bar{0}_V.$$

$$a_1, \dots, a_n = 0 \text{ by linear independence of } \beta$$

Isomorphisms

Proved that $T: V \rightarrow W$ invertible, β is basis of $W \rightarrow T(\beta)$ basis for $T(W)$

Corollary If $\dim(V) = n$ and $T: V \rightarrow W$ is invertible, then $\dim(W) = \dim(V) = n$

Def V and W are isomorphic if there is an invertible linear map $T: V \rightarrow W$. T is an isomorphism.

Isomorphic finite dimensional vector spaces

Thm if V is finite dimensional, then W is isomorphic to V iff $\dim W = \dim V$

Pf This is a corollary of the image of basis by invertible linear map, where if β is a basis for V , then $T(\beta)$ is a basis for W .

Show that if $\dim W = \dim V$, then they are isomorphic.

let α be the basis of W and β be the basis for V .

$$\alpha = \{w_1, \dots, w_n\} \quad \beta = \{v_1, \dots, v_n\}$$

Define $T: W \rightarrow V$ by $[T]_{\alpha}^{\beta} = I_n$.

$$T(w_i) = v_i$$

$$T(w_i) = T(a_1 w_1 + \dots + a_n w_n) = a_1 v_1 + \dots + a_n v_n$$

QED

Invertible Linear Map of A Matrix

$$\mathcal{L}(\mathbb{R}^n, \mathbb{R}^m) = \{L_A : A \in M_{m \times n}\}$$

For what A does L_A have an inverse?

$$L_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$\bar{x} \mapsto A\bar{x}$$

$$n = m.$$

There must be a map $(L_A)^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that

$$(L_A)^{-1} \circ L_A = I_{\mathbb{R}^n} = L_A \circ (L_A)^{-1} \quad \text{let } (L_A)^{-1} = L_B$$

$$L_B \circ L_A = I_{\mathbb{R}^n} = L_A \circ L_B$$

$$L_{BA} = L_{I_n} = L_{AB}$$

$$BA = I_n = AB$$

Def $A \in M_{n \times n}$ is invertible if $\exists B \in M_{n \times n}$ such that

$$BA = I_n = AB$$

Rmk The inverse of A (B) is unique if it exists

Let $BA = I_n = AB$ and $CA = I_n = AC$.

$$(CA)B = I_n(B) \nearrow$$

$$C(AB) = B = C(I_n)$$

$$C(I_n) = B \quad \boxed{C=B} \quad \text{QED}$$

Equivalent conditions for an invertible matrix

A is invertible $\Leftrightarrow L_A: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible

$\Leftrightarrow L_A$ is 1-1 and onto

$\Leftrightarrow N(L_A) = \{\vec{0}\}$

$\Leftrightarrow \{\vec{x} \mid A\vec{x} = \vec{0}\} = \{\vec{0}\}$

$\Leftrightarrow$ REF of A has leading entries in each row.

Matrix representation for an inverse linear transformation

Linear $T: V \rightarrow W$, β, γ is a finite basis for V, W

Claim if T is invertible, $[T^{-1}]_{\beta}^{\gamma} = ([T]_{\gamma}^{\beta})^{-1}$

$$\begin{aligned} [T^{-1}]_{\gamma}^{\beta} [T]_{\beta}^{\gamma} &= [T^{-1} \circ T]_{\beta}^{\beta} \\ &= [I_V]_{\beta}^{\beta} = I_n \end{aligned}$$

Finding an inverse matrix using RREF

$$AB = I_n$$

$$A(b_1 \dots b_n) = (e_1 \dots e_n)$$

$$Ab_1 = e_1, \dots, Ab_n = e_n \quad \Leftrightarrow \text{solve lin. sys. eq.}$$

$$(A \mid I_n)$$

$$(RREF(A) \mid B) \rightarrow \text{if } RREF(A) = I, B = A^{-1}$$

else A is not invertible.

generic matrix:

Claim $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible if $ad - bc \neq 0$, in which case

$$A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

proof by RREF

Change of coordinates

$[v]_{\beta}$ and $[v]_{\beta'}$, β and β' are finite bases

$$\begin{aligned} [v]_{\beta'} &= [I_V(v)]_{\beta'} \\ &= [I_V]_{\beta'}^{\beta} [v]_{\beta} \end{aligned}$$